Proposal 2

MPhil/PhD Proposal

Neurocognitive correlates of empathic and moral dysfunction in children with psychopathic antisocial behaviour

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Background:

Children with conduct problems (CP; i.e. antisocial behaviour) are a heterogeneous group. A particularly severe subtype are children with callous-unemotional traits (CP/CU+) who lack empathy and guilt and are at risk for developing adult psychopathy, a disorder characterised by moral dysfunction and persistent and severe antisocial behaviour (Frick & White, 2008). Children with CP/CU+ show a number of distinct cognitive, emotional and personality characteristics. They show abnormalities in their responsiveness to punishment cues, less trait anxiety, diminished responsiveness to distress cues in others and difficulties in distinguishing between moral (victim-based) and conventional (social order-based) transgressions (Frick & White, 2008).

It has been proposed that lack of empathy and immoral behaviour in individuals with CP/CU+ reflects lack of arousal to other people's distress as evidence by blunted response to distrss cues, such as fearful and sad expressions (Blair et al., 2005). It is hypothesised that the absence of a robust empathic responses to the distress cues of others leads to a failure to inhibit aggression towards them. According to Blair's model, moral socialization occurs through the pairing of the activation of the empathic response to distress cues with the representations of the act that caused the distress. Haidt's social intuitionist model (Haidt, 2001) also suggests that the ability to make quick, affective evaluations of actions is one of the key aspects of moral judgment and that to understand how human morality works research should be focused on the study of the emotional processes involved. On the other hand, other theoretical positions maintain that although emotion often accompanies moral processing, moral judgment of highly aversive actions may be independent of emotion (Huebner, Dwyer, & Hauser, 2008).

The amygdala and orbitofrontal cortex (OFC)/ventromedial prefrontal cortex (vmPFC) play a crucial role in empathic and moral processing (Blair, 2008), and have also been found to show diminished reactivity in children and adults with CP/CU+ during the performance of emotional-processing and morality tasks (Jones et al., 2009; Marsh et al., 2008; Finger et al., 2008; Glenn, Raine & Schug, 2009).

Although lack of empathy and moral dysfunction are hypothesised to predispose individuals to antisocial behaviour, there are no previous studies that 1) explore the relationship between empathy and morality using a wide battery experimental tasks and questionnaires probing empathic, affect and moral processing in typically developing children (TD); 2) provide a direct comparison of CP/CU+, CP/CU- (CP children without CU) and

TD children on a range of empathic, affect and moral processing measures; 3) assess the neurocognitive correlates of empathy and morality in CP/CU+, CP/CU- and TD children using brain imaging technology. This PhD thesis will address these gaps in knowledge focusing on boys, given the increased risk for antisocial behaviour in males. Details of the planned studies and specific hypotheses are delineated below.

Proposed PhD research:

Behavioural Study 1:

Objective: To study the relationship between empathic and moral processing in TD boys using a substantial battery of tasks and questionnaires.

Participants: 100 unmedicated, 10-16 year old boys recruited via existing school contacts.

Measures of Child Behaviour and Temperament: A range of established questionnaires will be given to obtain teacher/parent ratings of child behaviour and characteristics. These are *Strengths and Difficulties Questionnaire* (Goodman, 1997)*; Child and Adolescent Symptom Inventories (*4th Ed., Gadow & Sprafkin, 2002); and *Inventory of Callous-Unemotional Traits* (Frick, 2003).

Measures of Empathy and Affect: A range of established questionnaires and tasks will be used to assess empathic and empathy related processing. These will include *Griffith Empathy Measure (Dadds et al., 2007; parent-rated questionnaire)*; *Basic Empathy Scale* (Jolliffe & Farrington, 2007; self-rated questionnaire); a *Cartoon Empathy Task* recently developed for children (based on Vollm et al, 2006 and piloted in Dr Essi Viding lab); and Emotion Multimorph tasks (based on Calder et al. 1996 & Blair et al., 2001). In addition a novel empathy task assessing emotion attribution to self and others will be developed for this battery.

Measures of Moral Functioning: A range of established questionnaires and tasks will be used to assess moral processing. The prosocial scale of *Strengths and Difficulties Questionnaire* (Goodman, 1997; parent- and self-rated); The *Test of Self-Conscious Affect* (Tangney, 1996), a scenario-based measure of guilt and shame proneness; *The Moral/Conventional Task* (Turiel, 1983) a scenario-based measure of distinction between moral and conventional transgressions; *The Burnett Task of Social Emotions* (Burnett et al., 2008), a scenario-based measure of attributing moral and non-moral emotions to self and others; and the *Moral Sense Test* (http://moral.wjh.harvard.edu/; collaboration with Professor Marc Hauser at Harvard University), a scenario-based measure of moral judgment comprising moral personal and impersonal dilemmas.

Predictions: I expect to find modest to moderate correlations between the various empathy, affect and moral measures. I also expect that higher levels of CU correlate with lower levels of empathy and moral emotions as measured by our tasks and questionnaires.

In addiction, I want to explore whether higher levels of CU are also associated with differences in moral judgement, i.e. whether children with higher levels of CU are more likely to make utilitarian judgements for personal dilemmas.

Behavioural Study 2:

Objective: To provide a direct comparison of CP/CU+, CP/CU- (CP boys without CU) and TD boys on the above empathic and moral processing measures.

Participants: 20 boys scoring above 90^{th} centile for CP and CU; 20 boys scoring above 90^{th} centile for CP, but not CU; 20 TD boys. All groups will be matched for cognitive ability and age (range 10-16 years). The two CP groups will be matched for level of ADHD symptoms. All boys will be recruited via existing school contacts.

Measures: As in Study 1.

Predictions: I expect CP/CU+ boys to perform significantly worse than CP/CU- and TD boys on all empathic and moral processing measures, as well as on the aspects of affect measures assessing fear and sadness.

Neuroimaging Study:

Objective: To assess possible neural dysfunction associated with empathic and moral processing in CP/CU+.

Participants: 45 unmedicated, right-hand, males with ages between 10 and 16, with no history of substance abuse, divided in three groups of 15 each using the criteria described in Study 2.

Empathy Task: This task is based on an fMRI study by Vollm et al. (2006) and involves blocked presentations of cartoon stimuli of empathy evoking distress scenarios contrasted with theory of mind and physical causality scenarios. The task has been piloted for use with children.

Morality Task: This task would use an adaptation of the Moral Sense Test (http://moral.wjh.harvard.edu/; collaboration with Professor Marc Hauser at Harvard University), which involves short stories eliciting moral personal (emotion-provoking and involving salient harm to another person) and impersonal (less emotional) dilemmas.

Predictions: Empathy Task: I expect CP/CU+ boys to show attenuated empathic response to distress at the neural level, e.g. diminished amygdala reactivity, when compared with CP/CU- and TD boys.

Morality Task: In line with recent findings with adult psychopaths, I expect CP/CU+ boys to show less activation of the moral affect circuitry, e.g. diminished amygdala and mPFC reactivity, during the processing of aversive personal dilemmas, when compared with CP/CU- and TDC.

Selected Key References:

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